MATH 257
Linear Algebra with Computational Applications

Lectures and Discussions. For a detailed schedule of lectures, discussions, and their locations see Course Explorer:
https://courses.illinois.edu/schedule/2025/fall/MATH/257

- Lectures (2 hrs/wk): Mondays and Wednesdays
- Discussion Sections (1 hr/wk): Tuesdays and Thursdays at various campus locations
- TA office hours: Mon-Thur 5-7PM, see course LMS site for locations
- Instructor office hours: Berwick-Evans W3-4pm 314D Harker ; Chuang M4-5pm 233 CAB; [Note: students may attend the office hours of any instructor]

Teaching Staff.

- Lecture instructors: [PL1] Jer-Chin Chuang (jchuang@illinois.edu) and [PL2, PL3] Daniel Berwick-Evans (danbe@illinois.edu)
- Discussion: listed under the Course Information tab on the course LMS site (see below)

Learning Objectives.

- Familiarity with the core concepts of linear algebra
- Introduction to modern applications of linear algebra in a variety of fields
- Use of Python for linear algebra computations

Setup. This course consists of two hours per week of lecture and one hour per week of computing during the discussion sections.

Grading. Here is an executive summary of the grading scheme for this course, followed by drop and exemption modifiers, as well as percent-to-letter grade conversions and exam curving. Details about each category are given later in this document.

Grading categories

(1) 2% syllabus quiz
(2) 3% class attendance (checkpoint quiz for online section OL1 students)
(3) 10% discussion attendance / completion (33% / 67%)
(4) 8% PrairieLearn (non-lab) homework
(5) 5% PrairieLearn lab homework
(6) 6% Two Python quizzes (each 3%)
(7) 66% Three midterm exams (each 16% for total 48%) plus a Final Exam (18%)

Modifiers

- All assessments assigned in Weeks 1-2 (except the Syllabus Quiz due Friday Week 3) **will not count** towards your grade.
- For in-person sections, from Week 3 onwards, the lowest 3 attendance ~~daily iClicker~~ scores will be dropped.

- From Week 3 onwards, the two lowest weekly scores in each of the Items (3)-(5) above will be dropped.
- From Week 3 onwards, for lecture attendance the lowest three are dropped for in-person students, and for online students, the lowest six module checkpoint quizzes are dropped.
- If your Final Exam is higher than one of your midterm scores, it will replace the lowest midterm score. A missed midterm exam will be counted as receiving a score of 0%.
- Any assignments for which you are "exempted" will be dropped. See below for the requirements for being "exempted."
- If you were "exempted" for exactly one midterm, we will use the average of [the other two midterms + Final] as the score for the exam you were "exempted" from. Then, we will apply the midterm score replacement policy if applicable.
- If you were "exempted" for more than one midterm (*very* unlikely), contact your professor for grading details.

Special Cases

- Under certain special cases, the deadlines to assignments in categories (3)-(6) may be modified. See the "Absence Policy" document under *Course Information* on the course website.
- To be "exempted" from a midterm, you **need a letter from the Office of the Dean of Students** covering the *entire* exam window.
- There are no exemptions for the syllabus quiz or for the Final Exam.

Letter Grades

- ▶ 100.00 % - 98.00 % → A+
- ▶ 97.99 % - 93.00 % → A
- ▶ 92.99% - 90.00% → A-
- ▶ 89.99 % - 87.00 % → B+
- ▶ 86.99 % - 83.00 % → B
- ▶ 82.99 % - 80.00 % → B-
- ▶ 79.99 % - 77.00 % → C+
- ▶ 76.99 % - 73.00 % → C
- ▶ 72.99 % - 70.00 % → C-
- ▶ 69.99 % - 67.00 % → D+
- ▶ 66.99 % - 55.00 % → D
- ▶ 54.99 % - 0.00 % → F

Exam curves

We will curve each of the midterms and the Final Exam score such that the distribution of letter grades coincides with historic grade distributions for this course. No further curve will be applied at the end of the course.

In particular, there will be **no individual extra credit opportunities** after class instruction ends and closed assessments do not reopen. So make sure to work hard for every assessment!

Please check each week that your scores were entered correctly on Canvas. With so many students it can happen that your grade is entered incorrectly. If, after an

assessment, you find an error in the grading, please email the relevant course staff immediately. It can always happen that we made a mistake, so we always encourage you to see us if you think that happened. All rescoring requests must be made via email. Except for exams, rescoring requests will only be considered within a week of the assessment due date. For exams, rescoring requests will only be considered within a week of the posting of exam scores. Requests must be received before Reading Day. So don't wait!

With about 1200 students this semester there are always many cases where students are close (sometimes even very close) to the next letter grade, and at the end of the semester make the case that they should receive higher grades. Unfortunately, in almost all cases we can not grant the request without being unfair to other students—even if we would like to!

Introduction.

This is a first course in linear algebra. This covers basic definitions and algorithms of the subject needed in the higher level (engineering, science and economics) courses and more sophisticated mathematical techniques such as the Singular Value Decomposition.

In this course you learn the mathematical theory and how to implement it in Python. You will discover many of the striking modern applications of linear algebra, such as Google's PageRank algorithm, image and audio compression schemes such as JPEG and MP3, automatic face recognition and other data science and machine learning algorithms.

The course covers the same mathematical theory as MATH 415, but adds a focus on the computational and large data aspect of linear algebra through the lab sessions.

Technical equipment. Many aspects of this course will be conducted online. As such, each student will be assumed throughout the semester to have the necessary technical equipment to participate in course activities:

- a computer/laptop/tablet with a microphone,
- a stable internet access with sufficient bandwidth and data allowance for using Zoom
- ~~in-person section students will need to have iClicker access (either via app or remote)~~ use of iClickers during in-person lectures is optional

We do not grant accommodations due to technical issues accessing or completing assignments. Please contact the Student Assistance Center (helpdean@illinois.edu) immediately if you are missing any of required technology.

Other Linear Algebra courses. Be aware that course credit is not given for both MATH 257 and any of MATH 125, MATH 225, MATH 227, MATH 415, or ASRM 406. Any enrollment related questions should be sent to mathadvising@illinois.edu.

Three disclaimers.

- ⚠ This is not a course that only teaches you how to compute stuff. A computer will always be faster. Modern applications of linear algebra require a sophisticated understanding of theory and methods of linear algebra, and learning these is the purpose of this course. Some of it might look like abstract linear algebra. However, through the applications we cover in the discussions, you realize that this indeed is applied linear algebra.
- ⚠ If you already know some linear algebra, this course might look easy at the beginning. Don't be fooled into thinking it will stay like that. Even the material familiar to you will be covered in more depth here. Furthermore, the exams will require a deeper understanding of the concepts you already know something about. So it is a good idea to take this course seriously from the beginning.
- ⚠ If you find a typo or an error in any part of this course, please let us know by sending an email to the instructors. We appreciate your help, and are also happy to hear any further comments or suggestions. Thank you!

Course LMS site. The course site is as follows:

https://canvas.illinois.edu/courses/61184

All material will be available there or linked from there. Please note that if you have just registered for the course, you will automatically be given access within a few hours. Only if you do not have access to the course site 48 hours after registering, then contact your instructor.

Discussion section. Discussion sections are held in-person on Tuesdays and Thursdays with the detailed schedule available in Course Explorer:

https://courses.illinois.edu/schedule/2025/fall/MATH/257

Only attend the discussion section you are signed up for. No credit is given otherwise.

During discussion you will use computational tools in Python to solve linear algebra problems in real world applications in science and engineering. You will be working together in small groups on a Python Jupyter notebook. Though the first session is a Python tutorial, **students are assumed to have prior experience in Python or be able to familiarize themselves quickly to Python.** Students without the requisite programming background have struggled to complete the course and should consult their advisor. **Please bring your own device to the lab sessions.**

Attendance will be taken in the first 15 minutes via a code provided by the TA. You are responsible for entering the code in a timely manner. If there is an issue entering the code, your TA needs to be notified before the end of the session. Note that it is not enough to just be present. You have to be actively working with your group on the project. If this is not the case, we will not consider you present.

Requests to be excused should be submitted to the relevant teaching staff before the start of the session. Please note that due to the available drops in various assessment categories, we generally do not grant excused absences or extensions for short-term illnesses, but rather only in exceptional circumstances (e.g. university athletic participation or letter from Office of the Dean of Students).

Textbook. We will post extensive lecture notes for all lectures and practice problems online. For many students these notes are enough. If you still want to buy/download a book, here are some options (several of them free!) in no particular order:

- Philip N. Klein, Coding the Matrix: Linear Algebra through Applications to Computer Science, first edition, Newtonian Press
- Feryal Alayont, Steven Schlicker, Linear Algebra and Applications: An Inquiry-Based Approach, scholarworks.gvsu.edu/books/21/
- David Cherney, Tom Denton,Rohit Thomas, Andrew Waldron, Linear Algebra, www.math.ucdavis.edu/∼linear/
- Stephen Boyd and Lieven Vandenberghe, Introduction to Applied Linear Algebra - Vectors, Matrices, and Least Squares, https://web.stanford.edu/ boyd/vmls/
- Gilbert Strang, Linear Algebra and its Applications, fourth edition, Cengage.

You are not required to buy any of these textbooks. Please note the coverage and order of topics for these resources may differ from our course.

Slides and Videos. Lecture notes (slides) and module videos are available at the Canvas page of this course. An interactive version of the slides with fill-in boxes is also available. If you would like to use this feature, print out the fill-in slides and fill them out on your own or while watching the videos. Video errata may be found at the video link on the course site under the appropriate week's tab.
Additionally, each module has an associated module checkpoint quiz consisting of several conceptual questions with unlimited attempts. These quizzes are in PrairieLearn. **They are optional for students in the in-person sections but required for those in the online section OL1.** For OL1 students, the deadlines for these checkpoint quizzes are weekly.

Regardless, all are strongly encouraged to complete them as you study the modules. Be sure to spend some time thinking over the material in the module before attempting the quiz. Since these are conceptual questions, precision is important so read the lecture notes and the questions carefully. Most questsions are True/False but allow unlimited attempts.

There are many other great (free) videos about linear algebra. Here are some we recommend as an addition (not a substitute) for the lecture videos.

- Essence of Linear Algebra by 3Blue1Brown, on Youtube, highly recommended
- MIT lectures by Gilbert Strang, MIT Open Courseware
- Coding the Matrix videos by Philip Klein, on Youtube

In-person Lectures. In-person lectures are twice weekly on Mondays and Wednesdays. **You are expected to read the lecture slides and/or watch the lecture videos before attending the in-person lectures.** The content of the in-person lectures is not identical to that in the slides/videos. The in-person lecture will

highlight several of the concepts introduced in the slides/videos and provide further examples. Notes from the in-person lecture will be available at the course site the next day. Unless otherwise noted, students are responsible for all material presented in both the slides/videos (see above) and in-person lectures. **Attendance is mandatory for the in-person sections.** As with discussion sections, attendance will be taken in the first 15 minutes via Canvas/Moodle with a code provided by the instructors. During most in-person lectures there will also be some iClicker questions. You can either use the iClicker device or the iClicker app on your phone (available for Apple and Android smartphones here). Participation in the iClicker questions is optional, and the questions are ungraded.

Online Section. During the first two weeks students can move from any in-person section to the online section freely. Students can move from the online section to in-person sections subject to availability of seats in the in-person sections. Students in the online section will complete module checkpoint quizzes in lieu of attendance. (see subsection on Slides and Videos). All listed lecture instructors are instructors for the online section. Email about any issues that online students face should be copied to all listed lecture instructors. The subject line of the email must start with **[ONLINE]** written exactly as shown. Otherwise your email might be ignored.

Online homework. Each week there are two sets of homework (lab and non-lab) both delivered through PrairieLearn. These homeworks are for in-person and online sections. Both sets of homeworks are open-notes and you may collaborate with other students:

The non-lab homework associated with each module provides opportunity to practice the computations and algorithms covered in the module. In this homework you will have to do the computation we did in the video by yourself. The lab homework is based on the most recent Python lab activity in the discussion sections. See below for more information on PrairieLearn. This course is listed in PrairieLearn as follows:

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Workspace/help sessions. Starting Week 2, TAs will offer in-person tutoring Mondays to Thursdays 5-7pm. See the Office Hours tab on the course site for more details. At least one TA will be present during this time to answer your (non-Python related) questions. You don't have to come to the room only if you already have a question. You can just go there and work—either alone or with your fellow students. We will highly recommend that you form study groups and make use of this room!

For Python-related questions, undergraduate course assistants (CAs) will hold online office hours via Zoom throughout the week starting Week 2. See the course site for details.

Netiquette. Since this course has a substantial online component, please be respectful of your fellow classmates and teaching staff in all online communications. Fostering a helpful learning environment requires everyone's cooperation. Remember that forum posts are visible to all students and staff in the course (about 1200 people). So please double-check your posts before submitting them.

Weekly assignment schedule. Regular weekly assignments due dates are as follows:

- ⌛ PrairieLearn (non-lab) Homework (covering modules from previous week), due Tuesdays 11:59PM at 100% (see below)
- ⌛ PrairieLearn lab Homework (covering lab from previous week), due Thursdays 11:59PM.

For example, PrairieLearn Week 1 non-lab homework is due (at 100%) on Tuesday of Week 2.

Additionally, students participating in in-person sections are strongly encouraged to complete the optional, ungraded checkpoint quizzes (covering modules from current week). For students in the online section these checkpoint quizzes are mandatory and graded by completion.

PrairieLearn. We will use PrairieLearn for homework. See the Quick links at the top of the course page.

(Non-lab) Homework (at 100%) will be due on Tuesdays at 11:59PM. The first homework is due on Tuesday of Week 2. The PraireLearn homework will lean towards computations, while the checkpoint quizzes will focus more on conceptual problems.

How points are given on PrairieLearn. PrairieLearn places emphasis on mastery. The idea is to keep doing questions until you master the underlying concept or method. Once you do, you should be able to answer these questions very quickly.

The way this works in PrairieLearn is that each question has a value, a point total, and a point maximum. If you answer a question correctly, two things happen:

- ▶ The point total increases by the value, until you reach the point maximum.
- ▶ The value increases.

If you answer a question incorrectly, one thing happens:

- ▶ The value goes back to what it was originally.

This system rewards repeated correct answers, which tend to demonstrate mastery. There is no penalty (other than resetting the value) for answering a question incorrectly, so don't be afraid to submit an answer. Similarly, don't be afraid to keep doing a question after you reach the point maximum - your point total with never go down!

Credit. There is no need to "submit" your homework. The system will record whatever your score is at that time. However, you'll note the following line at the top of your screen:

Available credit: 110% until 11:59PM, Fri, 08/29

What this means is that if you reach 100% prior to 11:59PM on that Friday - i.e., complete the homework early - you will receive an extra 10% bonus. You will see this reflected in your score (the instant you reach 100%, it will jump to 110%).

If you click on the "?" just to the right of the line about available credit, you'll see all the dates associated with this homework. In particular, it says:

- ▶ you can receive 100% until 11:59PM, Tuesday, 09/02,
- ▶ you can receive 80% until 11:59PM, Tuesday, 09/09.

Note that your score will never go down. For example, if you achieve 90% by 11:59PM on Tuesday, 09/02, you won't be able to increase your score after that time, but you won't be penalized for not reaching 100% - your score will remain 90% forever. On the other hand, if you achieve only 70% by 11:59PM on Tuesday, 09/02, you will be able to increase your score after that time (to a maximum of 80%).

Lab Homework (at 110%) will be due on Thursdays at 11:59PM. The first lab homework is due on Thursday of Week 2. Unlike the non-lab homework, the lab homework will only be offered at 110% and 80%.

Any changes to regular homework deadlines for either type will be announced on CampusWire.

Please note that your overall PrairieLearn score in either lab/non-lab homework categories is capped at 100%. So even if you score 110% on every assignment, you will only receive 100% overall. The bonus is designed to help off-set homeworks where you may not have received full credit.

Typos/Errors. If you believe there is a typo or an error in a question, or if you believe your answer was graded incorrectly, please take a screenshot and post to CampusWire. We have access to all of your submissions and can check to see what, if anything, went wrong.

CampusWire. All announcements will be posted on CampusWire at

https://campuswire.com/p/G41FB50C9

Please make sure you are signed up for CampusWire. The registration PIN is available under the Quick links at the top of the course site. When posting on CampusWire, please use the subject line wisely and post in the appropriate category. For example, if you ask something about matrix multiplication in Lecture notes 5, write "Lecture notes 5 - Matrix multiplication" and not just "Question about matrices". In addition, please post to the entire class whenever this is appropriate. No question will ever be held against you.

Because of the large number of students (about 1200) and the limited number of teaching staff, please help us facilitate response times by observing the following:

- Please post all content questions and those of general interest regarding course material or organization to CampusWire rather than via email. Multiple course staff monitor CampusWire, and this allows all students to benefit from both the question and our reply.
- For private questions regarding special cases of course policy (e.g. DRES, absence from an exam, etc.) please use email and copy the relevant teaching staff (e.g. TA, lab TA, or course instructor). Please clearly indicate your lecture/discussion/lab sections in the email subject. We aim to have response times for emails generally in one or two business days.

Syllabus Quiz. Because of the online format for much of this course, familiarity with course policies will be essential. All students will be required to complete by **11:59PM Friday of Week 3 (September 12)** a syllabus quiz at the course LMS site under the "Course Information" tab. This quiz covers basic course policies. It is open-notes and unlimited attempts are allowed. Please complete this assessment as early as possible.

Exams. This course will have two Python quizzes, three midterms, and a final exam, all administered by CBTF.

Students register with CBTF for an exam slot during the below specified **tentative** windows. Details regarding administration of the final exam will be announced later in the semester.

- **Midterm 1: Week 4, 09/15-09/17**
- **Midterm 2: Week 8, 10/13-10/15**
- **Python Quiz 1: Week 10, 10/27-10/29**
- **Midterm 3: Week 12, 11/10-11/12**
- **Python Quiz 2: Week 14, 12/04-12/07** (Note Thursday-Sunday window)
- **Final: 12/11-12/16**

Students are responsible to ensure they register sufficiently early for availability of seats, are properly registered with CBTF, and sit for their exam at their registered time. There will be neither make-up exams nor extensions due to seat unavailability. Requests to be exempted from a midterm exam require an absence letter from the Office of the Dean of Students that explicitly states your absence over the entire relevant exam window.

CBTF. This course uses the College of Engineering Computer-Based Testing Facility service CBTF Online for its exams:

https://cbtf.illinois.edu/index.asp

The policies of the CBTF are the policies of this course, and academic integrity infractions related to the CBTF are infractions in this course.

If you have accommodations identified by the Division of Rehabilitation-Education Services (DRES) for exams, please upload your Letter of Accommodations (LOA) to the CBTF site: https://cbtf.illinois.edu/students/dres before you make your first exam reservation (at least five business days prior).

If you have any issue during an exam, please inform the proctor or relevant CBTF staff immediately. Work with the proctor to resolve the issue at the time before logging off, and ask your proctor to file an incident report as record of your interaction. If you do not inform a proctor of a problem during the test, then you forfeit all rights to redress.

Review all instructions on the CBTF website before your first exam:

https://cbtf.illinois.edu/students/rules-copy

Cheating. No books, notes, cheat sheets or electronic devices are allowed during the exams except a non-graphing scientific calculator. We take cheating very seriously! A more detailed description of the University policy on cheating and plagiarism may be found in the following link:

http://www.las.illinois.edu/students/integrity/

Date	Week	Lecture	Lab	Topic
08/25	1	1		Intro to linear systems, Matrices
08/27	2	2		Echelon form of matrices, Gaussian Elimination
	1		1	Python tutorial
09/01				No class (Labor Day)
09/03	2	3		Linear combinations, Matrix-vector & Matrix multiplication
	2		2	Working with vectors
09/08	3	4		Properties of matrix multiplication, Elementary matrices
09/10	3	5		Matrix inverses and their computation
	3		3	Matrix operations
09/15	4	6		LU decomposition, Solving linear systems using LU
09/17	4	7		Mass-spring system, Inner products
09/15-09/17	4			Midterm 1
09/22	5	8		Subspaces of $\mathbb{R}^n$, Column spaces and nullspaces
09/24	5	9		Abstract vector spaces, Linear independence
	5		4	Solving systems of linear equations
09/29	6	10		Basis and dimension, The four fundamental subspaces
10/01	6	11		Orthogonal complements, Graphs
	6		5	Graphs and Algebraic Graph Theory
10/06	7	12		Coordinates, Orthonormal bases
13/08	7	13		Linear transformations, Coordinate matrix
	8		6	Data compression
10/13	8	14		Determinants, Cofactor expansion
10/15	8	15		Eigenvectors and Eigenvalues and their computation
10/13-10/15	8			Midterm 2
10/20	9	16		Properties of eigenvectors, Markov matrices
10/22	9	17		Diagonalization
	9		7	Markov Chains
10/27	10	18		Powers of matrices, Matrix Exponential
10/29	10	19		Linear Differential Equations
10/27-10/29	10			Python Quiz 1
	10		8	Dynamical Systems
11/03	11	20		Orthogonal projection onto lines and subspaces
11/05	11	21		Least squares solutions, Linear Regression
	11		9	Linear Regression
11/10	12	22		Gram-Schmidt process and QR decomposition
11/12	12	23		Spectral Theorem
11/10-11/12	12			Midterm 3
11/17	13	24		SVD
11/19	13	25		Low rank approximations, Pseudo-Inverse
	13		10	SVD and applications
11/24-11/28				No class (Fall break)
12/01	14	26		Principal Component Analysis
12/03	14	27		Review complex numbers, Complex linear algebra
	14		11	Principal Component Analysis
12/04-12/07	14			Python Quiz 2
12/08-12/10				No lectures (offset Python quizzes)